Effect of Orbital Altitude on Satellite Revisit Time over Central Europe

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This study compares geometric Earth-observation access over Central Europe for four initially circular satellite orbits at nominal altitudes of 400, 500, 600, and 700 km. Thirty-day GMAT trajectories are analysed over an 800-point geographic grid using an idealised nadir-pointing sensor with a fixed 20-degree half-angle. The regional area-weighted mean of per-target mean revisit gaps is 42.86, 28.51, 29.96, and 26.30 h, respectively. Average daily coverage increases from 46.98% at 400 km to 68.48% at 700 km. However, the largest observed complete gap is shortest at 600 km, at 88.20 h, compared with 111.45 h at 700 km and 420.12 h at 500 km. These differences show that mean performance and worst observed interruption favour different altitudes. Geometry checks and a 20-second coarsening diagnostic support initial assessment of the analysis, but do not establish fine-step convergence. The results are preliminary, configuration-specific estimates; finer temporal and geographic sampling and additional initial-geometry cases are required before a final mission-design recommendation.

I. Introduction

Earth-observation mission design requires a relationship to be established between the information sought on the ground and the opportunities available from orbit. For a regional monitoring mission, the ability to observe a location once is insufficient to describe its usefulness: the time between successive opportunities is also relevant. Revisit performance therefore provides a means of comparing orbital configurations against a temporal observation objective. Coverage studies must nevertheless distinguish an opportunity to view a target from the acquisition of a usable image. The former follows from orbital and sensor geometry, whereas the latter may also depend on illumination, cloud cover, instrument performance, and operational scheduling.

Professional mission studies treat coverage as a coupled orbit–instrument problem. Nag et al. [1] explore Earth-imaging constellation architectures using orbital and instrument variables, evaluate temporal coverage metrics, and compare results with an independent analysis system. Their work provides a methodological precedent for defining design variables, computing geographic access, and examining differences between implementations. A complementary lesson is that simulation output requires explicit supporting evidence. Hughes et al. [2] describe the verification and validation of GMAT using independent reference tools and comparison tests. These examples motivate a study in which performance comparisons and checks on the analysis method are developed together.

The present study narrows the design space to one satellite and four candidate initial altitudes. Increasing altitude changes the orbital period and, for a fixed angular field of view, the sensor footprint. The resulting revisit performance also depends on the ground-track pattern, target latitude, and selected orbital orientation. A comparison based only on the number of daily revolutions would therefore omit the geometry that determines whether a target is observable. The research question is how altitude affects mean and maximum geometric revisit gaps over Central Europe when nominal inclination and sensor angular field of view are held fixed. The contribution is a reproducible regional comparison and an explicit account of its numerical limitations, rather than a new coverage algorithm.

The completed baseline combines GMAT propagation with Python processing of time-tagged Earth-fixed spacecraft positions. All cases use the same 30-day interval, target grid, and sensor model. Alongside regional mean performance, the comparison examines the largest complete gap and its geographic distribution. A coarsening diagnostic evaluates the effect of retaining every other trajectory sample; finer-step and initial-geometry campaigns remain future work. Dividing the computation between a propagation tool and an analysis driver follows an established engineering workflow: Nickel et al. [3], for example, describe automated GMAT–MATLAB simulations for the TESS mission. Cloud cover, imaging task scheduling, and downlink capacity remain outside the present baseline.

Section II defines the observation study. Section III presents the geometry and metrics. Section IV reports completed checks and outstanding verification work. Section V discusses the numerical comparison, followed by the conclusions in Section VI.

II. Study Definition

A. Region and Target Grid

The region of interest is a user-defined rectangle spanning 5°–25° east longitude and 45°–55° north geodetic latitude. This explicit boundary is used as a practical Central European study area, rather than as an official geographic definition. Both land and water within the rectangle are retained. Country boundaries provide map context only.

The rectangle is divided into $0.5° \times 0.5°$ cells. Their centres define 40 longitude values and 20 latitude values, giving 800 targets. Longitudes run from 5.25° to 24.75° east and latitudes from 45.25° to 54.75° north. Targets lie at zero height on the WGS 84 ellipsoid. Figure 1 shows the grid. A cell is represented by its centre point; observing the point does not imply that the whole cell has been imaged.

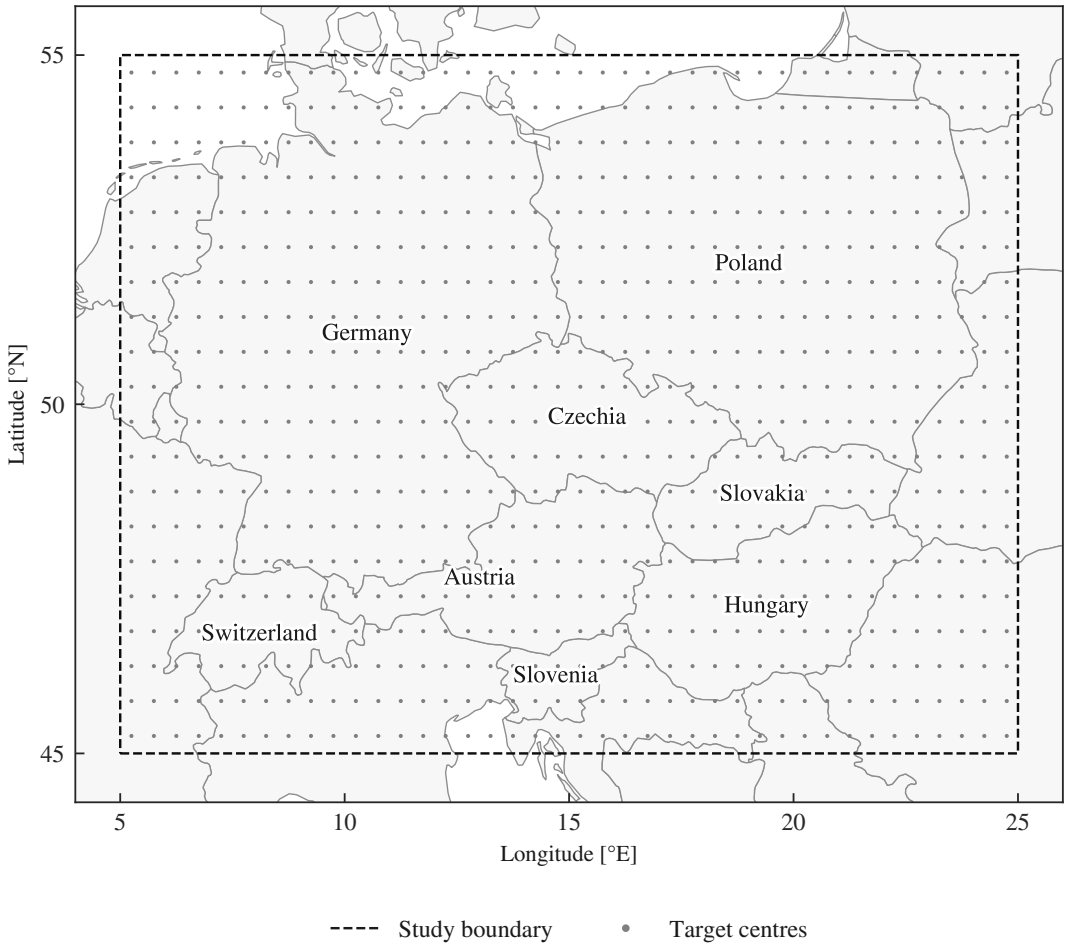


Fig. 1 Rectangular study region and the 800 target locations. The grid includes land and water. Country boundaries: Natural Earth.

B. Orbit Cases and Observation Concept

Four trajectories were generated for nominal initial altitudes of 400, 500, 600, and 700 km. The common setup specifies an initially circular orbit at 60° inclination, with right ascension of the ascending node, argument of perigee, and true anomaly set to zero in the initial EarthMJ2000Eq representation. For a circular orbit, the sum of the last two angles defines the initial argument of latitude. No sun-synchronous condition is imposed.

The initial semi-major axes are 6778.137, 6878.137, 6978.137, and 7078.137 km, using 6378.137 km as the reference radius for the altitude labels. These labels describe the initial configuration, rather than a constant height above an

oblate Earth throughout propagation. The common interval extends from 1 January 2026 at 00:00 UTC to 31 January 2026 at 00:00 UTC.

An idealised circular conical sensor points towards Earth's centre, with half-angle $\alpha = 20°$ and full angular field of view 40°. There is no off-nadir steering. Its boresight is geocentric nadir, which is distinct from the local ellipsoid normal. Geometric access requires a target to lie inside the cone and to have an unobstructed line of sight. No daylight or cloud filter is applied, so night-time opportunities are included. The sensor is a modelling assumption, not a selected instrument.

Table 1 Baseline analysis definition and trajectory-input record.

Parameter	Value
Nominal initial altitude	400, 500, 600, 700 km
Nominal initial eccentricity / inclination	0 / 60°
Nominal initial RAAN / argument of latitude	0° / 0°
Common UTC interval	1–31 January 2026, 30 days
Exported position frame / units	GMAT EarthFixed / km
Samples per trajectory	259,201, including both endpoints
Output spacing	Approximately 10 s
Sensor half-angle / pointing	20° / towards Earth's centre
Target geometry	Zero-height WGS 84 ellipsoid
Region / target count	5–25°E, 45–55°N / 800
Illumination, clouds, scheduling	Not modelled

The analysis verifies timestamps, frames and initial position radii directly from the CSVs. The complete initial-state and force-model definitions must also be retained in the original GMAT scripts for full reproducibility; the CSV exports alone do not identify the force model or integrator.

III. Analysis Method

Figure 2 links the common observation assumptions and the four trajectories to the reported coverage and revisit measures. The calculation is applied independently to each altitude with identical target and sensor definitions.

A. Trajectory Inputs and Earth Geometry

The input record contains elapsed seconds, A1 modified Julian epoch, A1 Gregorian date, and spacecraft EarthFixed Cartesian coordinates. All four files contain matching sample times from 0 to 2,592,000 s. Absolute A1 epochs are retained for consistency checks; daily bins are formed from elapsed seconds referenced to the common UTC start. A1 dates are not interpreted as UTC dates.

Ground coordinates are converted to Earth-fixed Cartesian positions using WGS 84 semi-major axis $a_E = 6378.137$ km and flattening $f = 1/298.257223563$, as specified by NGA [4]. With $e_E^2 = f(2 - f)$ and geodetic latitude and longitude (ϕ_j, λ_j),

$$N_j = \frac{a_E}{\sqrt{1 - e_E^2 \sin^2 \phi_j}}, \qquad \mathbf{g}_j = \begin{bmatrix} N_j \cos\phi_j \cos\lambda_j \\ N_j \cos\phi_j \sin\lambda_j \\ N_j(1 - e_E^2) \sin\phi_j \end{bmatrix}. \tag{1}$$

Here $\mathbf{g}_j$ is the zero-height target position. The outward unit normal is

$$\mathbf{n}_j = \begin{bmatrix} \cos\phi_j \cos\lambda_j \\ \cos\phi_j \sin\lambda_j \\ \sin\phi_j \end{bmatrix}. \tag{2}$$

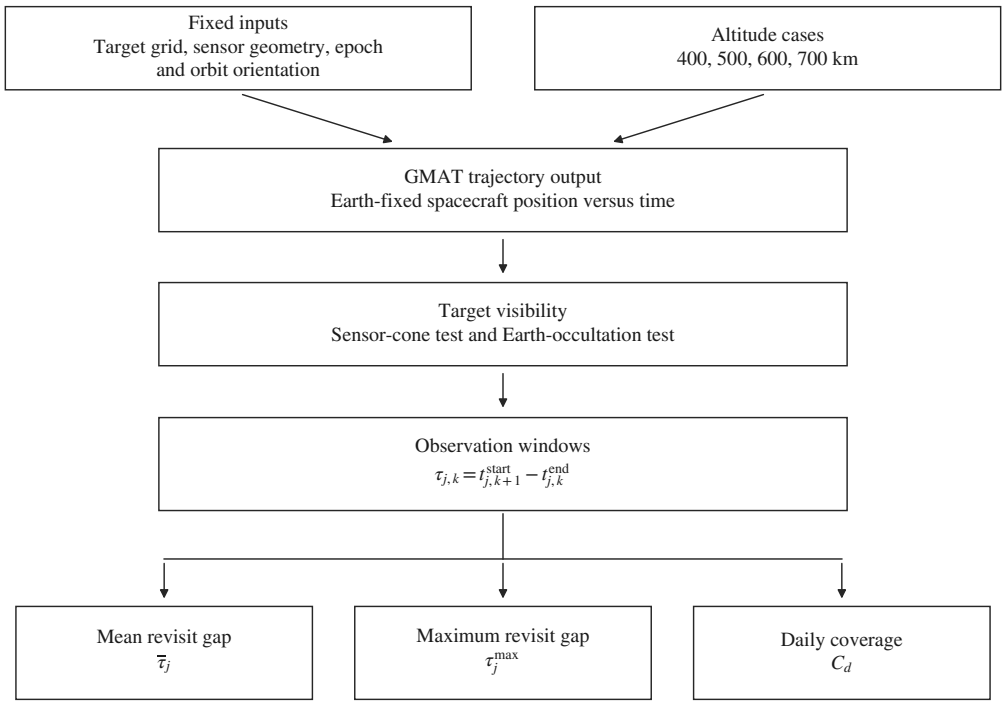


Fig. 2 Analysis structure linking altitude-dependent trajectories to observation windows and performance measures. The equation defines revisit gap as the interval from the end of one observation window to the start of the next.

B. Visibility and Observation Windows

For spacecraft position $\mathbf{r}(t)$, the sensor boresight and spacecraft-to-target unit vector are

$$\mathbf{b}(t) = -\frac{\mathbf{r}(t)}{\|\mathbf{r}(t)\|}, \qquad \mathbf{u}_j(t) = \frac{\mathbf{g}_j - \mathbf{r}(t)}{\|\mathbf{g}_j - \mathbf{r}(t)\|}. \tag{3}$$

A sample is visible when both conditions hold:

$$\mathbf{b}(t) \cdot \mathbf{u}_j(t) \geq \cos\alpha, \qquad [\mathbf{r}(t) - \mathbf{g}_j] \cdot \mathbf{n}_j > 0. \tag{4}$$

The first condition applies the sensor cone. The second is the horizon condition for a target on a convex ellipsoid, excluding lines of sight through Earth. Terrain and atmospheric refraction are omitted.

Consecutive visible samples are grouped into observation windows. At an internal transition, the window boundary is estimated at the midpoint between the visible and invisible sample times. A window touching the analysis endpoint is clipped there and flagged. This procedure approximates boundary timing; it does not solve continuous-time access events and can miss a sufficiently short window between output samples.

For target j, with K_j windows, the complete gap between windows k and $k+1$ is

$$\tau_{j,k} = t^{\text{start}}_{j,k+1} - t^{\text{end}}_{j,k}, \qquad k = 1, \ldots, K_j - 1. \tag{5}$$

Leading and trailing unobserved portions of the 30-day interval are boundary-censored. They are stored separately and excluded from the set of complete gaps. A target with fewer than two windows receives no mean or maximum complete-gap estimate, rather than a zero value.

C. Revisit and Coverage Metrics

For targets with at least two windows, the mean and maximum complete gaps are

$$\bar{\tau}_j = \frac{1}{K_j - 1} \sum_{k=1}^{K_j - 1} \tau_{j,k}, \qquad \tau^{\max}_j = \max_k \tau_{j,k}. \tag{6}$$

Regional summaries use approximate spherical cell-area weights. For a cell with longitude width $\Delta\lambda_j$ and latitude edges ϕ^-_j, ϕ^+_j,

$$\widetilde{w}_j = \Delta\lambda_j(\sin\phi^+_j - \sin\phi^-_j), \qquad w_j = \frac{\widetilde{w}_j}{\sum_l \widetilde{w}_l}. \tag{7}$$

Angular quantities in this expression are in radians. These weights compensate for unequal latitude–longitude cell areas, but are not exact ellipsoidal areas. The regional mean is

$$\bar{\tau}_R = \sum_j w_j \bar{\tau}_j, \qquad \tau^{\max}_R = \max_j \tau^{\max}_j. \tag{8}$$

Equation (8) averages per-target means, not all individual gaps pooled across the region. All 800 targets have complete gaps in the baseline. If estimates are missing, the implementation renormalises the mean over available targets and separately reports the excluded area.

Daily coverage C_d is the weighted fraction of targets with a nonzero-duration observation window overlapping day d:

$$C_d = 100 \sum_j w_j I_{j,d}, \qquad \bar{C} = \frac{1}{30} \sum_{d=1}^{30} C_d. \tag{9}$$

Here $I_{j,d}$ is one if such a window exists and zero otherwise. Each day is a 24-hour bin from the common start epoch. Coverage refers to sampled target points and is an approximation of observed regional area.

D. Implementation and Reproducibility

The Python calculation evaluates visibility in chunks, groups samples by target, and exports per-target statistics, observation-window tables, daily coverage, and regional summaries. The recorded run uses Python 3.12.14, NumPy 2.3.5, and Matplotlib 3.10.8; GeoPandas is used for country-boundary plots. The metadata records input SHA-256 hashes, the observation model, software versions, and trajectory checks. Natural Earth boundaries are used for cartographic context [5].

IV. Verification and Sampling Diagnostic

A. Completed Input and Geometry Checks

All trajectories span the same 30 days, with 259,201 finite samples, strictly increasing elapsed times, and matching absolute initial epochs. Their initial Earth-centred radii agree with the four nominal semi-major axes in Table 1. Exported variables are spacecraft positions in EarthFixed, rather than Earth's position in that frame.

The vectorised visibility test was compared with a direct implementation of the boresight/line-of-sight dot product and surface-normal horizon test. Known nadir and antipodal cases were checked. An analytic equatorial footprint boundary supplied a further test: on the equator the WGS 84 section is a circle, and, for an off-nadir half-angle below the apparent limb, the footprint half-width in geocentric angle is

$$\psi = \sin^{-1}\left(\frac{r}{a_E}\sin\alpha\right) - \alpha. \tag{10}$$

Targets just inside and outside that boundary were classified as expected. An additional direct-vector comparison at 1,000 distributed epochs for the largest-gap 500 km target agreed with the vectorised result. These checks address implementation consistency; they do not independently validate the full propagated mission or establish access-event timing accuracy.

B. Ten-Second Versus Twenty-Second Sampling

The baseline uses the approximately 10-second output. A second calculation retains every other sample, producing a 20-second series without rerunning GMAT. Table 2 and Fig. 3 report the changes. The regional mean gap increases by 0.78–2.89%, and mean daily coverage falls by 0.36–1.15 percentage points. At 400 km, the largest observed gap increases from 172.79 to 183.02 h, a 5.92% change.

Table 2 Coarsening diagnostic. Changes are relative to the 10-second baseline.

Altitude (km)	Mean-gap change (%)	Coverage change (pp)	20 s largest gap (h)
400	+2.89	-1.15	183.02
500	+1.21	-0.62	420.12
600	+1.25	-0.56	88.21
700	+0.78	-0.36	111.45

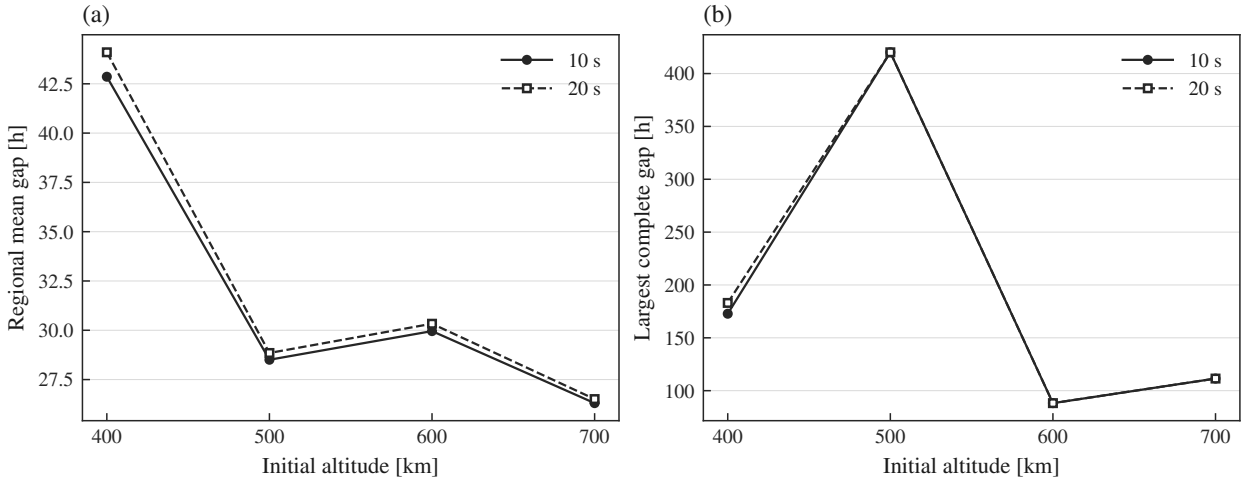


Fig. 3 Revisit metrics from 10-second samples and a 20-second subsample. This is a coarsening diagnostic, not evidence of convergence towards a finer-step solution.

The regional-mean and largest-gap altitude rankings are unchanged under this coarsening. However, short windows may be missed at both resolutions, and stable rankings do not bound numerical error. Native GMAT output at 5 s or finer, a finer target grid, extended durations, and different initial orbital orientations have not yet been analysed.

V. Results and Discussion

A. Regional Performance and Altitude Trade-Off

Table 3 summarises the 10-second baseline. All 800 targets have at least two observation windows in every altitude case, so the mean includes the whole sampled region. The area-weighted mean gap falls from 42.86 h at 400 km to 26.30 h at 700 km, a 38.62% reduction. The trend is not monotonic: the 600 km mean gap, 29.96 h, exceeds the 500 km value of 28.51 h. Average daily coverage increases monotonically across these four cases, from 46.98% to 68.48%, an increase of 21.50 percentage points.

Table 3 Preliminary 30-day geometric-access results at approximately 10-second sampling.

Altitude (km)	Regional mean gap (h)	Largest complete gap (h)	Mean daily coverage (%)	Min. daily coverage (%)
400	42.86	172.79	46.98	43.18
500	28.51	420.12	54.36	50.76
600	29.96	88.20	61.67	55.24
700	26.30	111.45	68.48	55.33

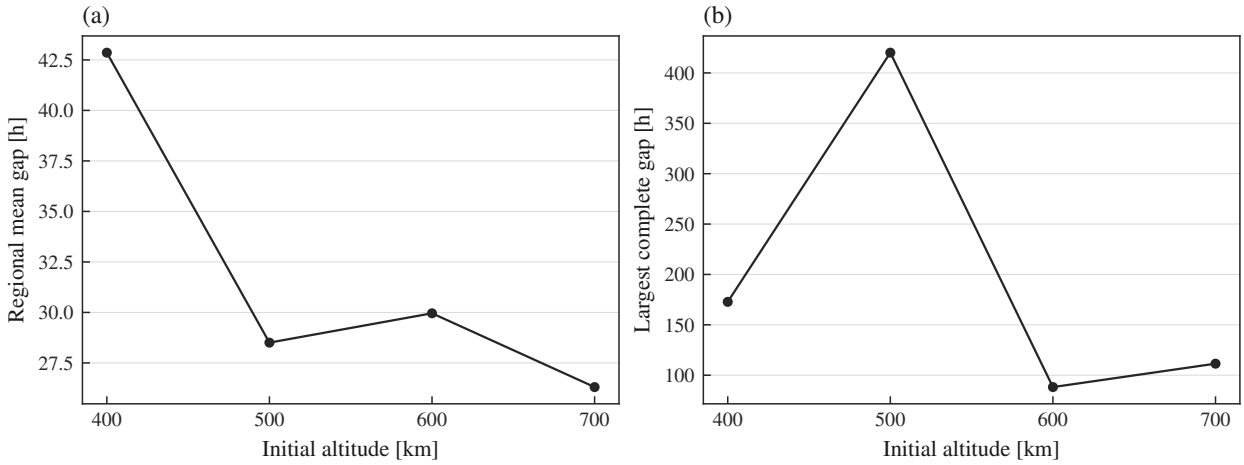


Fig. 4 Regional mean of per-target mean gaps (left) and largest observed complete gap across the grid (right). The two measures favour different altitudes.

The 700 km case is strongest on regional mean gap and average daily coverage, whereas 600 km gives the smallest largest observed gap, 88.20 h. The 500 km case has the longest complete interruption, 420.12 h (approximately 17.51 days), despite its relatively favourable regional mean. Thus a selection based only on the regional mean would omit a material aspect of service continuity. The reported maxima are the longest complete gaps observed in this interval, not guaranteed bounds on future performance.

B. Geographic Distribution and Long-Gap Locations

Figure 5 maps the per-target mean gaps with one common scale. At 400 km, mean gaps generally increase towards the southern part of the region. The 700 km distribution is lower overall, while the 500 km case shows pronounced

spatial structure. These results support reporting geographic detail alongside a single regional average.

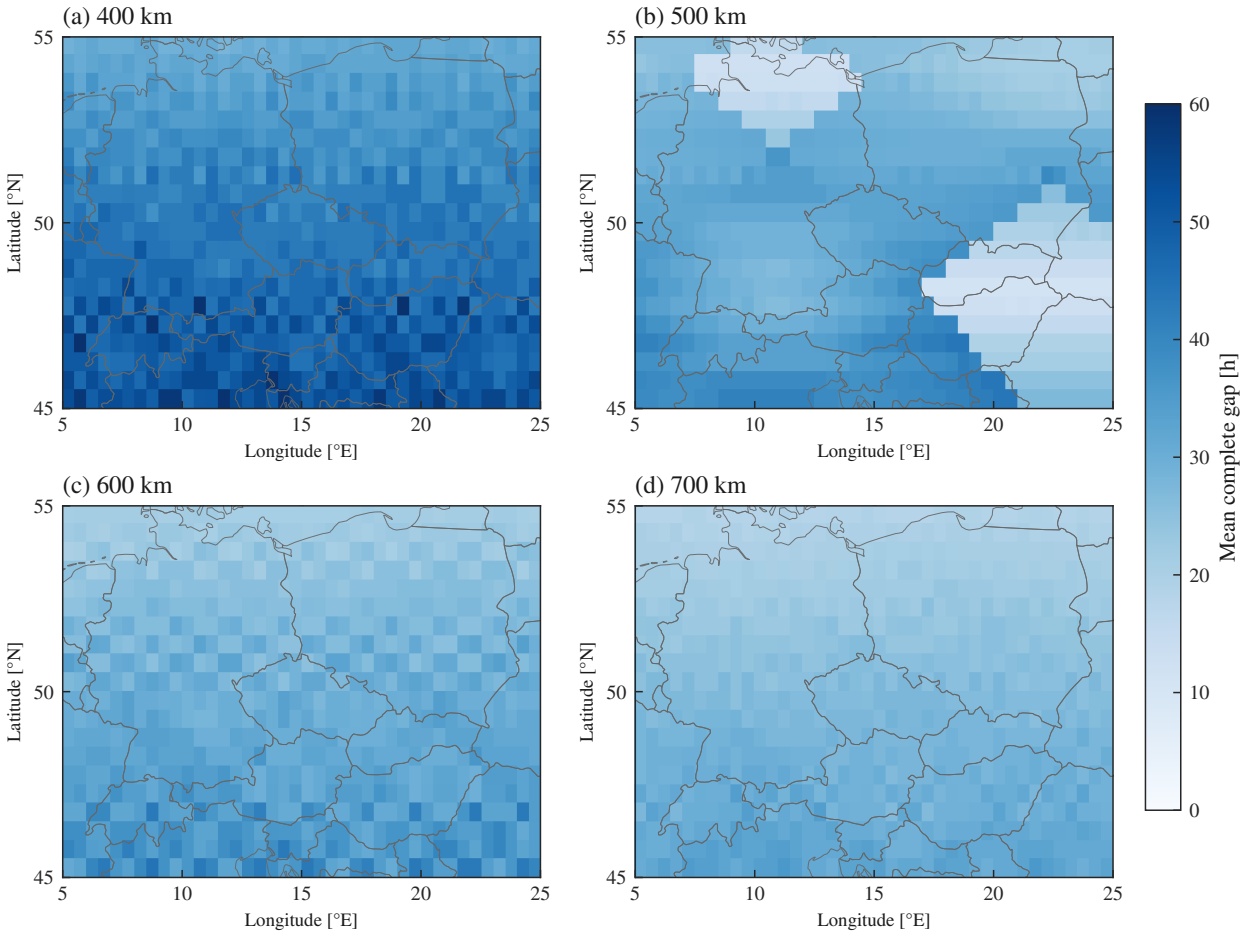


Fig. 5 Per-target mean complete revisit gap. The same colour scale is used in all panels. Each cell is represented by its centre target; country boundaries are supplied by Natural Earth.

The maximum-gap maps in Fig. 6 reveal an additional pattern. The 500 km long gaps affect a spatial band, rather than appearing solely as one isolated target. One largest-gap target is at 11.75°E, 48.25°N; its 26 detected windows include the 420.12 h complete gap. The spatial structure is compatible with altitude-dependent pass geometry, but a specific resonance or repeat-cycle explanation has not been tested. The extreme requires finer temporal sampling and inspection of its access history before it is treated as a robust design result.

C. Daily Coverage and Distribution of Mean Gaps

Figure 7 shows how coverage changes from day to day. The 700 km case has the highest 30-day average, but its lowest daily value is 55.33%, close to the 600 km minimum of 55.24%. An average coverage advantage therefore does not imply an equally large advantage on the least favourable day. None of the configurations provides full regional coverage on every day of the sampled interval. This is consistent with the restricted single-satellite observation concept, and is not a statement about the performance of a constellation or a steerable instrument.

Figure 8 presents the cumulative area-weighted distribution of per-target mean gaps. The 700 km curve lies largely to the left of the others, while 500 and 600 km trade relative performance across different parts of the distribution. This distribution complements the regional means; it does not describe the probability of acquiring usable imagery.

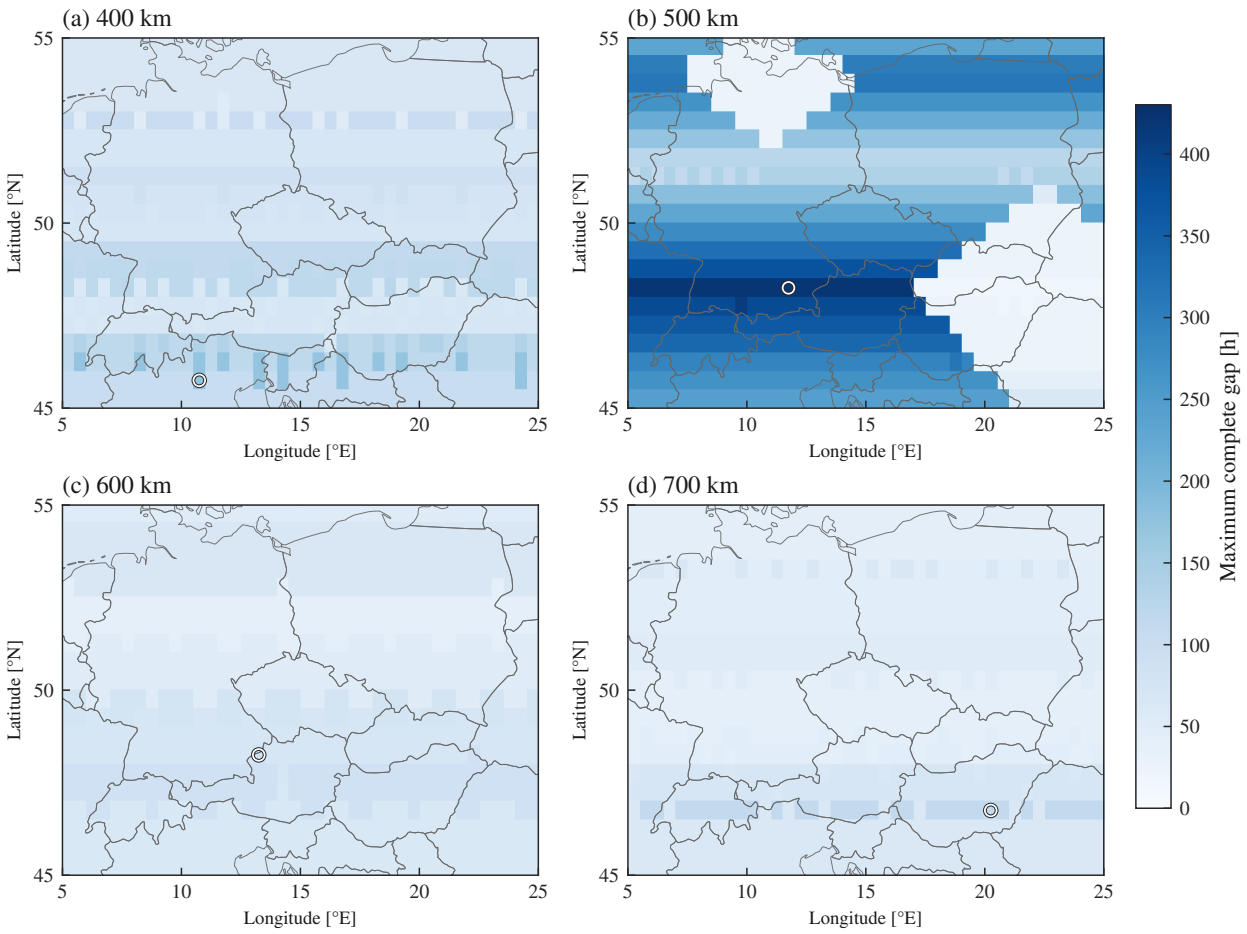


Fig. 6 Largest observed complete revisit gap at each target. Darker colours indicate longer interruptions; all panels use a shared scale. White circles identify one largest-gap target per altitude. Boundary-censored intervals are excluded. Country boundaries: Natural Earth.

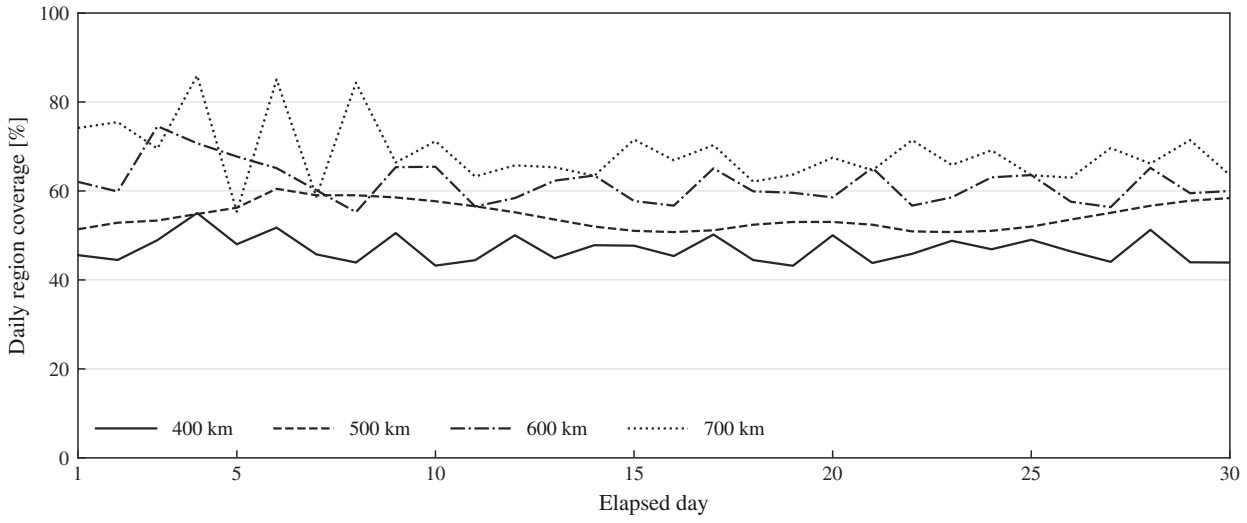


Fig. 7 Daily area-weighted fraction of targets with at least one geometric opportunity. Each bin spans 24 hours from the common UTC start; cloud cover, illumination and image scheduling are excluded.

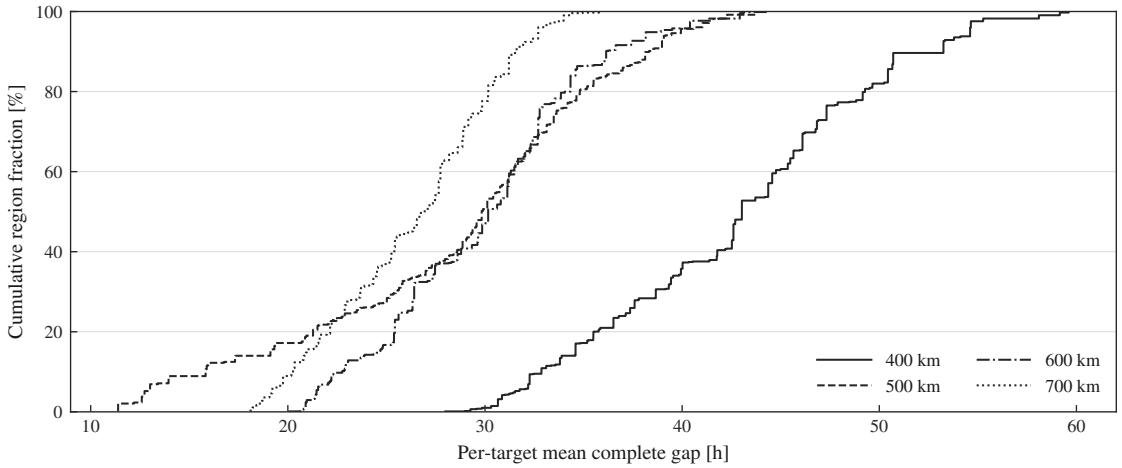


Fig. 8 Cumulative region fraction as a function of per-target mean complete gap. Area weights refer to the full target region; all baseline targets have defined complete-gap estimates.

D. Limitations and Remaining Work

The results depend on the initial epoch and orbital geometry, a fixed 20-degree sensor half-angle, and the chosen regional boundary. The sensor has no steering or imaging-quality constraint. Coverage includes night-time and cloudy conditions, and says nothing about data return, ground sampling distance, battery capacity, or an operational imaging schedule.

Temporal sampling can miss short windows, while midpoint boundary estimates introduce event-time approximation. Geographic sampling represents cells by points and has not been refined. The finite interval censors leading and trailing unobserved periods; excluding them can affect comparisons when windows cluster in time. A long observed complete gap is therefore useful evidence about the sampled scenario, but is not a mission-lifetime maximum. The spherical area weights also approximate ellipsoidal cell areas.

Finally, the exported CSVs do not preserve GMAT's gravity-field truncation, perturbation selections, integrator settings, or software release. The intended setup used a simplified Earth-gravity model including J_2, with other perturbations disabled, but these settings have not been audited against the native scripts. Archiving and checking the four original GMAT scripts is required before presenting the study as fully reproducible. These limitations should be resolved through a controlled verification campaign, rather than by interpreting more decimal places in the existing summary statistics.

VI. Conclusion

For the specified single-satellite, 20-degree half-angle observation concept, nominal altitude changes materially affect regional geometric access. Across the four sampled configurations, 700 km gives the lowest regional area-weighted mean complete gap, 26.30 h, and highest average daily coverage, 68.48%. Relative to 400 km, these represent a 38.62% reduction in mean gap and a 21.50 percentage-point increase in average daily coverage. Nevertheless, mean revisit performance does not improve monotonically with altitude.

A different criterion leads to a different preferred case: 600 km has the shortest largest observed complete gap, 88.20 h, compared with 111.45 h at 700 km. The 500 km case demonstrates the limitation of average-only selection, combining a regional mean of 28.51 h with a 420.12 h interruption at a target in the region. A mission decision must therefore specify whether typical access, geographic fairness, or avoidance of long gaps is the controlling requirement. None of these results establishes guaranteed daily access to the whole region.

The analysis supports a preliminary trade study, not a universal optimum. Input checks and independent geometry calculations passed, and the 10-to-20-second coarsening retained the altitude rankings; however, fine-step and grid convergence are unproven. The next steps are to archive the complete GMAT configurations, rerun with finer output sampling, refine the target grid, and test initial geometry and duration. Until those checks are complete, 700 km is the leading candidate for mean access and coverage, and 600 km for the observed worst-gap criterion, within this scenario only.

A. Supporting Access Visualisations

Figure 9 shows the first-day subsatellite tracks. Its latitude is geocentric, unlike the geodetic target-grid latitude, and it is used only for qualitative orientation. Sensor footprints are not shown. It does not replace the visibility calculation in Eq. (4).

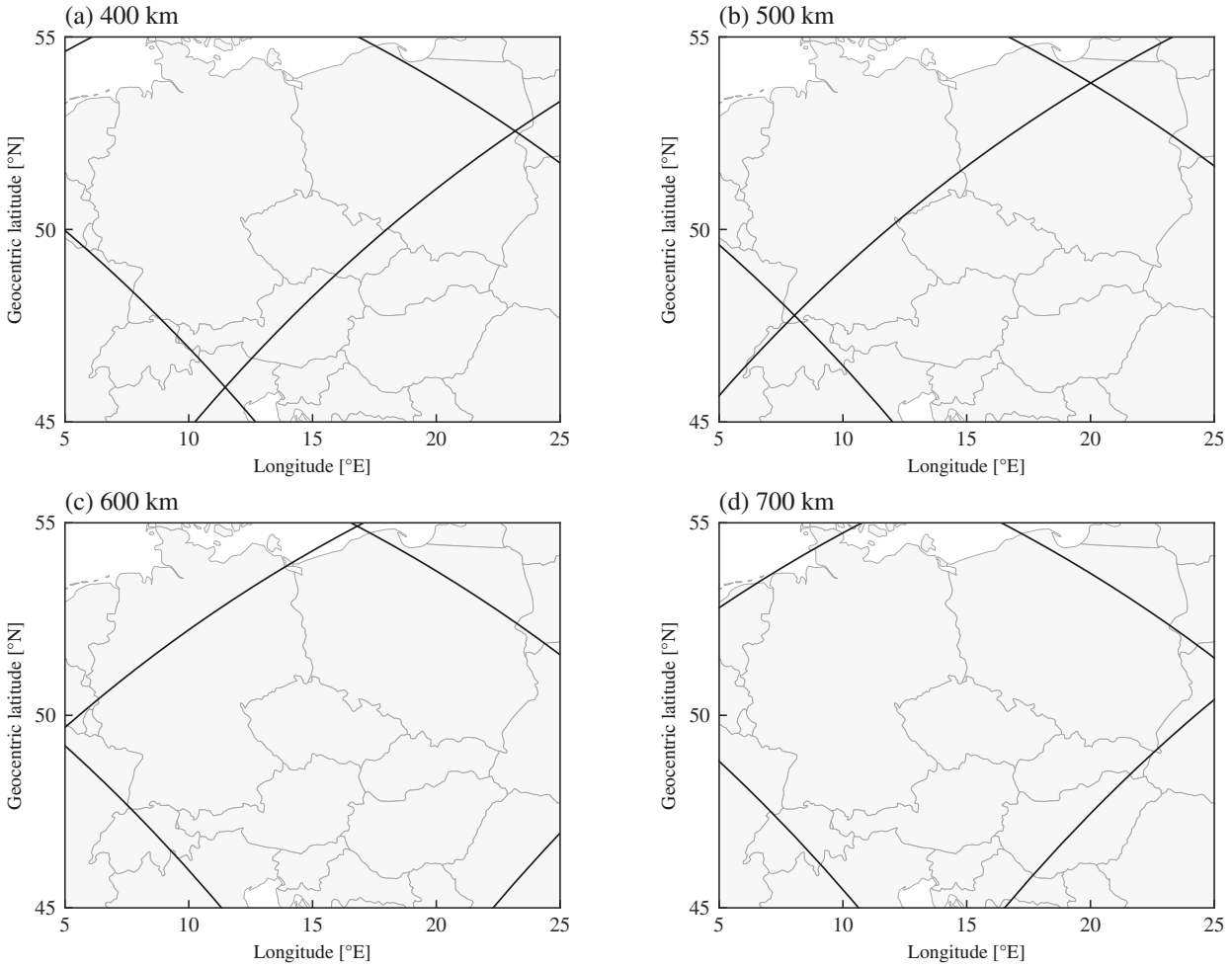


Fig. 9 First-day subsatellite tracks, shown with geocentric latitude. Country boundaries provide context; these lines do not indicate the full sensor footprint. Country boundaries: Natural Earth.

Figure 10 illustrates the first 72 hours for the grid point at 14.75°E, 49.75°N. Markers show window centres to keep short windows visible at this scale; bars retain their sampled durations. This example is illustrative and is not the 500 km largest-gap target.

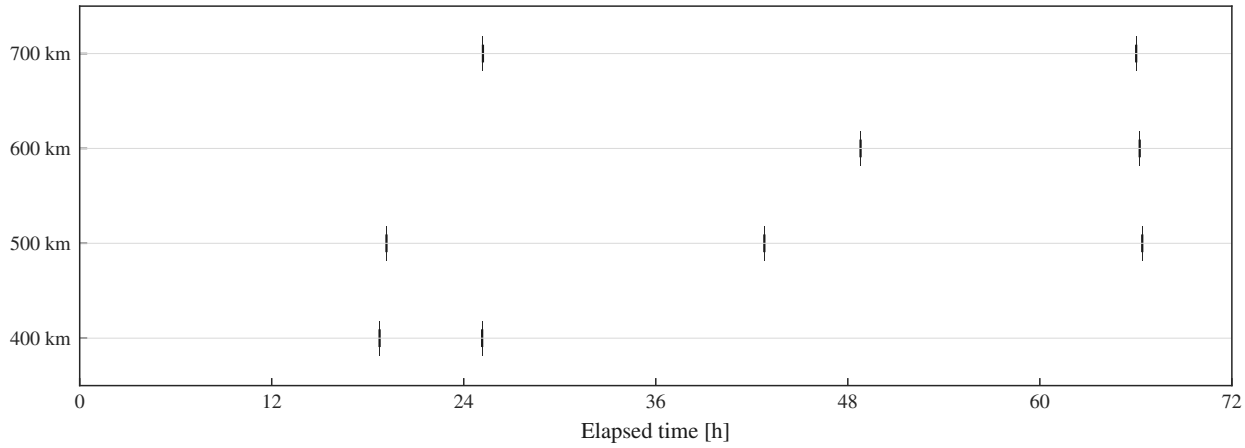


Fig. 10 Observation windows for an example target during the first 72 hours. Markers indicate window centres; their marker widths do not represent observation duration.

References

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